

DRTNet: Diffusion Reconstruction Texture Network for AI-generated Image Detection

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Abstract—Diffusion models have achieved significant success in AI-generated images, making it challenging for human to distinguish between synthetic and real images. Therefore, there has raised concerns about the potential misuse with malicious intent. The existing methods struggle with detecting unseen generated images and exhibit limited robustness against unseen post-processing procedures. To address these issues, we introduce a novel generated-image detector called Diffusion Reconstruction Texture Network (DRTNet), which leverages diffusion reconstruction errors from Stable Diffusion and employs texture feature extraction. Specifically, DRTNet incorporates a dual-branch input that includes both the input image and the reconstructed image to gather comprehensive information. Then the noise patterns, neighboring pixel relationship, and frequency artifacts are extracted from the samples. The extracted features of the input image and the reconstructed image are fused by dual cross-attention to consider the inter-dependencies between the dual-branch inputs. DRTNet offers an effective method for detecting images generated by unseen diffusion models or GANs. Notably, the proposed DRTNet demonstrates remarkable robustness against unseen post-processing procedures. Extensive experimental results show that DRTNet outperforms the state-of-the-art generated-image detectors.

Index Terms—AI-generated image detection, diffusion model, diffusion reconstruction, texture feature extraction.

I. INTRODUCTION

The advent of the Denoising Diffusion Probability Model (DDPM) [1] has revolutionized the field of image generation due to its unparalleled proficiency in producing high-quality images. Currently, diffusion model has emerged as the dominant methodology in text-to-image generation, such as Stable Diffusion [2], DALL-E2 [3], Midjourney, *etc.* Despite the powerful generative capabilities of diffusion models, there is a growing concern about their potential misuse such as malicious deepfake, fraudulent schemes, AI-assisted plagiarism, *etc.* Therefore, the development of a robust detector specifically tailored to discriminate diffusion-generated images has become an imperative necessity.

The generation processes of diffusion model differs entirely from that of Generative Adversarial Networks (GAN), therefore the traces that can be used to detect GAN generated images, for example, Fourier domain checkerboard artifacts, are difficult to detect for diffusion model generated images [4].

This work is supported by the National Natural Science Foundation of China under Project 62201600, NUDT Research Project ZK22-56, and NUDT Innovation Science Foundation 23-ZZCXKXKY-07.

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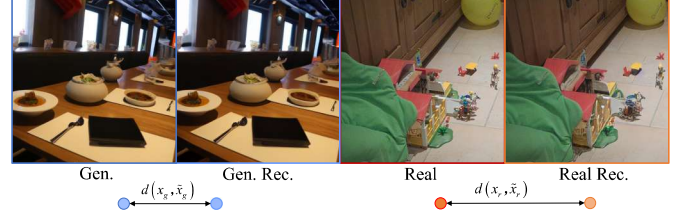


Fig. 1. Illustration of the reconstruction error of the generated sample and the real sample through the Stable Diffusion's VAE. The reconstruction error of the real sample is significantly larger than that of the generated sample.

It is a common practice to train a CNN binary classifier [5], [6] on real and generated images. However, this approach suffers from limited generalization ability and robustness to unseen images. Diffusion reconstruction [7]–[9] provides an alternative solution, which evaluates the reconstruction error between the input image and its reconstructed image using a pre-trained generative model. It leads to an improved robustness of the detectors.

In this paper, we focus on developing a generalized and robust AI-generated image detector. We propose a **Diffusion Reconstruction Texture Network (DRTNet)**, which considers the differences in reconstruction error, high-frequency components, inconsistencies among neighboring pixels, and frequency artifacts of AI-generated images. DRTNet consists of a Diffusion Reconstruction (DR) module, a Texture Feature Extraction (TFE) module, and a Feature Fusion (FF) module. We adopt the Stable Diffusion to reconstruct the input image, which maps the input image to the latent space and then returns to image space. The underlying assumption of diffusion reconstruction is that images generated through diffusion processes can be more precisely reconstructed using a pre-trained diffusion model than real ones, as illustrated in Fig. 1. Then DRTNet takes the input image and its reconstructed image as a dual-branch input, which provides more texture information. The TFE module analyzes the noise patterns obtained by Space Rich Model (SRM) [10], RGB features obtained by Neighboring Pixel Relationship (NPR) module, and frequency features adaptively obtained by an Invertible Neural Network (INN) [11]–[13] for the input image and its reconstructed image. And then a dual cross-attention is utilized to fuse the features of the dual-branch image samples. Subsequently, the prediction layers distinguish between the generated images

and the real ones by using the extracted features. Extensive experiments validate that DRTNet effectively improves generalization ability. The proposed method exhibits high detection accuracy on unseen diffusion models and GANs, and robustness against various unseen perturbations.

The main contributions are summarized as follows:

- We propose DRTNet, a diffusion reconstruction texture network, for detecting generated images which employs the input images and their reconstructed images through diffusion model's VAE as a dual-branch inputs to provide comprehensive texture information.
- DRTNet extracts texture features including noise patterns, neighboring pixel relationship, and frequency features and utilizes a dual cross-attention mechanism to fuse the features of the input image and the reconstructed image.
- Extensive experimental results demonstrate that DRTNet exhibits superior generalization ability and robustness in detecting unseen generated images.

The rest of the paper is organized as follows: Section II introduces the related works in AI image generation and detection. Section III presents the design details of DRTNet. Section IV introduces the implementation details and the experimental results. Finally, Section V concludes the paper.

II. RELATED WORK

A. AI Image Generation

Generative Adversarial Networks (GANs) [14] and Variational Autoencoders (VAEs) [15] have long been at the forefront of AI image generation. Nevertheless, the emergence of DDPM [1] has challenged their dominance, establishing a new paradigm. Consequently, a number of methods have been proposed to refine the architectures, accelerate the sampling process, and etc. Denoising Diffusion Implicit Models (DDIM) [16] is introduced to reduce the number of sampling steps of DDPM using non-Markovian diffusion processes. Liu *et al.* [17] consider DDPM as a method for solving differential equations on manifolds and propose Pseudo Numerical Diffusion Models (PNDM). Subsequently, a large number of diffusion models for text-to-image generation have been developed. Another notable work is Latent Diffusion Models (LDM) [2], which introduces the idea to perform diffusion on latent space and conditions diffusion models using a cross-attention mechanism. Stable Diffusion [2] build upon LDM and further refined to deliver substantial improvements in generation performance.

B. AI-generated Image Detection

The detection of AI-generated images has attracted increasing attention. Traditional methods primarily leverage spatial features and frequency artifacts. Frequency in Face Forgery Network (F3Net) [5] utilizes the frequency-aware forgery clues. Xi *et al.* [6] introduce a dual-stream network (DualNet) comprising a residual and a content stream. The former employs the Spatial Rich Model (SRM) [10] for texture extraction, while the latter captures low-frequency traces. Training deep learning classifiers to directly detect generated images

often suffers from poor generalization ability and struggles to effectively detect unseen generation models. CNNDet [18] utilizes a pre-trained ResNet50 [19] as a binary classifier to examine whether there are common patterns in the Fourier domain of GAN-generated images. Corvi *et al.* [4] verify that frequency artifacts in diffusion models are less noticeable compared to those in GAN models.

Recent diffusion-generated detectors leverage diffusion reconstruction [7]–[9], [20] or CLIP embeddings [21]. Notably, diffusion model generated images are more easily reconstructed by pre-trained diffusion models compared with real images. Diffusion Reconstruction Error (DIRE) [7] employs DDIM to measure the error between an image and its reconstruction. However, DIRE is limited by inefficient sampling and requires a lengthy training duration. Chen *et al.* present diffusion reconstruction contrastive training (DRCT) [8] towards universal detection of diffusion-generated images. AEROBLADE [20] adopts a training-free strategy for latent diffusion images using autoencoder reconstruction errors. Though AEROBLADE proves highly effective with high-definition pixel images, it has reduced effectiveness when applied on low-resolution images. Meanwhile, George *et al.* [9] propose FakeInversion which detects unseen text-to-images utilizing the inversion of Stable Diffusion.

III. PROPOSED METHOD

In this paper, we propose a Diffusion Reconstruction Texture Network (DRTNet) which is designed to enhance the generalization ability and robustness of the detectors by analyzing textures inherent in both input image and its reconstructed image, including noise patterns, neighboring pixel relationship in the RGB channels, and frequency artifacts.

A. Overview of DRTNet Framework

The overview of DRTNet is illustrated in Fig. 2. It comprises three main modules, *i.e.*, a Diffusion reconstruction module, a Texture Feature Extraction (TFE) module and a Feature Fusion (FF) module. Given an input image $\mathbf{x}$, the reconstructed image $\tilde{\mathbf{x}}$ is obtained using a pre-trained VAE encoder and decoder of Stable Diffusion [2]. Then, the TFE module extracts features from both the spatial domain and frequency domain, including the noise patterns captured using high-pass filters of SRM [10], the color features are derived from the RGB channels, and frequency features obtained through an INN with DCT. In the FF module, a dual cross-attention mechanism is then employed to adaptively assign weights to the fused features of the input image and the reconstructed image. And a cross-entropy loss function is introduced to guide the classifier to learn an optimized feature representation.

B. Diffusion Reconstruction Module

We adopt a dual-branch input approach, taking both the input image $\mathbf{x}$ and its reconstructed image $\tilde{\mathbf{x}}$ as input. The reconstruction process involves inverting the input image $\mathbf{x}$ through the diffusion model's VAE to obtain the reconstructed image $\tilde{\mathbf{x}}$. During the forward process, the input image $\mathbf{x}$

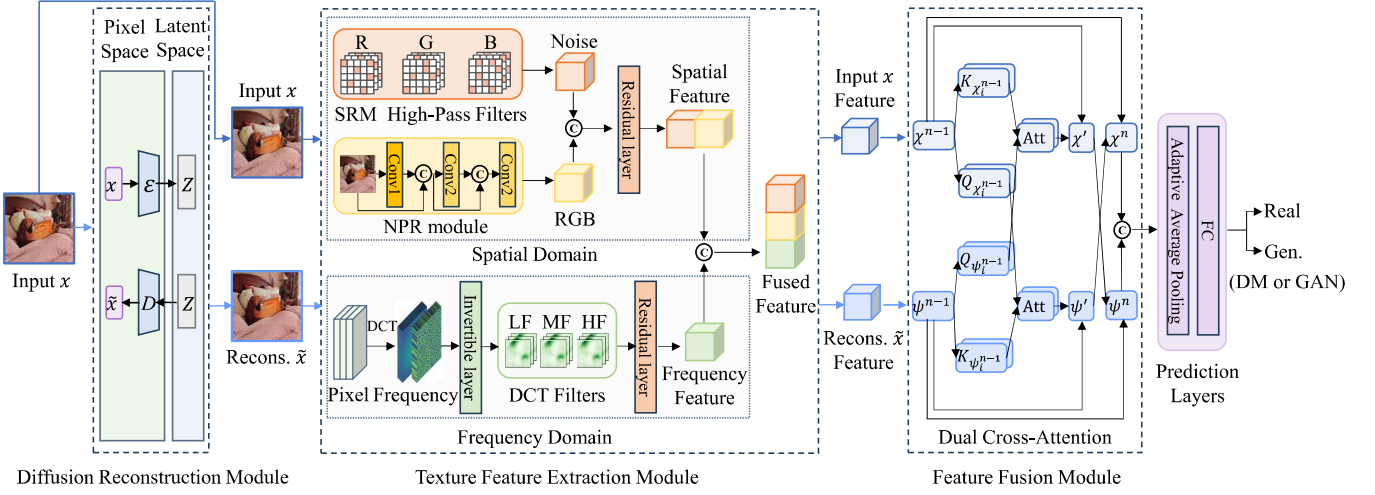


Fig. 2. Overview of the **Diffusion Reconstruction Texture Network (DRTNet)** framework. In the diffusion reconstruction module, the image is mapped to the latent space with Stable Diffusion’s VAE encoder, and then it is mapped back to pixel space using Stable Diffusion’s VAE decoder. The input image and reconstructed image are used as dual-branch inputs. In the Texture Feature Extraction (TFE) module, the features including noise patterns, neighboring pixel relationship, and frequency features are extracted from the input image and its reconstructed image. In the Feature Fusion (FF) module, the features of the input image and the reconstructed image are fused using a dual cross-attention mechanism.

is initially mapped to the latent space $\mathbf{z} = \mathcal{E}(\mathbf{x})$ with diffusion model’s VAE encoder $\mathcal{E}(\cdot)$. In the reverse process, the reconstruction $\tilde{\mathbf{x}} = \mathcal{D}(\mathbf{z})$ is mapped back to image space by employing diffusion model’s VAE decoder $\mathcal{D}(\cdot)$. The diffusion-reconstructed image $\tilde{\mathbf{x}}$ can then be expressed as:

$$\tilde{\mathbf{x}} = \mathcal{D}(\mathbf{z}) = \mathcal{D}(\mathcal{E}(\mathbf{x})). \quad (1)$$

The reconstruction-based detection is built upon two main assumptions: (i) given a generative model G_i and a generated image $\mathbf{x}_g$, we can obtain a reconstructed version $\tilde{\mathbf{x}}_g = \mathcal{D}(\mathcal{E}(\mathbf{x}_g))$ utilizing the reconstruction method derived from G_i , (ii) the reconstructed distance $d(\mathbf{x}_g, \tilde{\mathbf{x}}_g)$ between the generated image and its reconstructed version is small. In contrast, a real image $\mathbf{x}_r$ cannot be reconstructed accurately, that is:

$$d(\mathbf{x}_r, \tilde{\mathbf{x}}_r) > d(\mathbf{x}_g, \tilde{\mathbf{x}}_g). \quad (2)$$

Based on the above assumptions, diffusion model’s VAE is more effective at reconstructing generated images compared to real images. An intuitive explanation is that the VAE utilizes a latent representation $\mathbf{z}$ that maps to a constrained data manifold [20]. Real images typically contain rich details and complex texture information, and there may be inconsistencies between neighboring pixels, which may not conform to the preset probability distribution between pixels in generated images. Generated images reside within this manifold and can thus be accurately reconstructed, whereas real images are mapped to the nearest point on the manifold, resulting in higher reconstruction errors. Consequently, the dual-branch inputs, which include both the input images $\mathbf{x}$ and the reconstructed images $\tilde{\mathbf{x}}$, provide more comprehensive information for feature extraction in texture networks. Moreover, the Stable Diffusion’s VAE reconstruction accelerates the reconstruction process, eliminating the computationally substantial deterministic denoising process.

C. Texture Feature Extraction Module

In the Texture Feature Extraction (TFE) module, the Spatial Rich Model (SRM) [10] noise, neighboring pixel relationship, and frequency features of both the input and the reconstructed images $\mathbf{x}$, $\tilde{\mathbf{x}}$ are captured. It has been empirically confirmed that generated images tend to leave artifacts specially in the high-frequency components [4] due to the up-sampling operations in the image generation process. The neighboring pixels [22] and frequency feature [5] in generated images may exhibit inconsistencies. Therefore, we utilize SRM to obtain the noise patterns, which extracts the high-frequency components of an image through a series of convolutional high-pass filters. The Neighboring Pixel Relationship (NPR) module extracts RGB features by performing convolution on the input image and concatenating the output f_1 with the image $\mathbf{x}$, resulting in a 6-channel color feature map. This output f_1 is then passed through a convolution with adjustable parameters to produce f_2 . Both f_1 and f_2 are concatenated and subsequently convolved to obtain the final color feature map. These two types of spatial features are concatenated along the channel dimension. To comprehensively extract the frequency features of an image, Invertible Neural Network (INN) with DCT is utilized to analyze its low, medium, and high frequency content. DCT decomposes an image into its constituent frequency components, thereby enabling detection of the frequency-based artifacts inherent in generative models. To facilitate a learnable and adaptive DCT feature extraction process, we employ an INN layer designed to learn the frequency features. Lastly, the spatial and frequency domain features are integrated by concatenating them channel-wise.

Given an image $\mathbf{x}$, the feature extraction process using DRTNet is visually depicted in Fig. 3. The left sub-images are the noise patterns of the R, G, and B channels obtained through

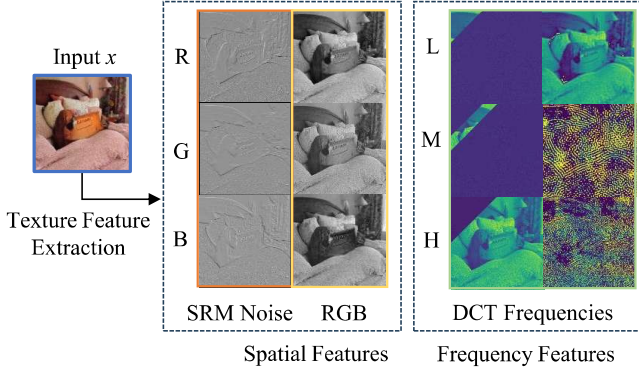


Fig. 3. An example of the texture feature extraction. The left sub-images are noise pattern obtained by SRM for the R, G, and B channels. The middle sub-images are color correlation diagrams obtained by NPR. The right sub-images are low, medium, and high frequencies obtained by INN with DCT.

SRM. The middle sub-images analyze the color consistencies through NPR. The right sub-images display the low, medium, and high frequencies obtained by INN with DCT.

D. Feature Fusion Module

In the Feature Fusion (FF) module, a dual cross-attention mechanism is employed to assign weights to the fused features of the input image $\mathbf{x}$ and the reconstructed image $\tilde{\mathbf{x}}$. By integrating the dual-branch features, the network enhances the inter-dependencies between both the input image $\mathbf{x}$ and the reconstructed image $\tilde{\mathbf{x}}$, thereby improving its detection capabilities, providing more accurate and robust results.

First, the output of the input image feature χ and the reconstructed image feature ψ are flattened and normalized. Then, a cross-attention mechanism is employed to better fuse the input image feature χ and the reconstructed image feature ψ . A Dual Cross-Attention (DCA) relies on three primary elements: the Q (Query), K (Key), and V (Value). The DCA assesses the correlation degree among distinct vectors at each spatial position. The input image features χ acts as the Q, and attention weights (K/V) are computed based on the reconstructed image features ψ , highlighting inter-dependencies between the dual-branch inputs. Given multi-heads h , we split χ^{n-1} , and ψ^{n-1} into h segments, obtaining $\chi_i^{n-1}, \psi_i^{n-1} \in \mathbb{R}$, $i \in \{1, 2, \dots, h\}$. Within each head i , the process yields a Query vector $\{Q_{\chi_i^{n-1}}, Q_{\psi_i^{n-1}}\} \in \mathbb{R}$ and Key vector $\{K_{\chi_i^{n-1}}, K_{\psi_i^{n-1}}\} \in \mathbb{R}$ by linear projection respectively. Then χ'_i is obtained by multiplying the attention matrix with χ_i^{n-1} . d_k represents the dimension of a token.

$$\chi'_i = \text{softmax}\left(\frac{Q_{\psi_i^{n-1}} K_{\chi_i^{n-1}}^T}{\sqrt{d_k}}\right) \chi_i^{n-1}. \quad (3)$$

After obtaining χ'_i from each head, they are concatenated to form χ' , and is then added to ψ^{n-1} to produce the output ψ^n . Similarly, χ^n can be obtained. Subsequently, a layer normalization is applied, followed by an MLP module. The FF module adopts a transformer encoder structure.

TABLE I
THE DESCRIPTION OF TESTING DATASETS.

Dataset	Resolution	Test Number	Source	Seen
Real	256×256	2k	ALASKA	✓
DALL-E2 [3]	256×256	2k	T2I [6]	✓
SDv1 [2]	512×512	1k	ImageNet	✗
ADM [23]	256×256	1k	ImageNet	✗
LDM [2]	256×256	1k	LSUN Bed.	✗
PNDM [17]	256×256	1k	LSUN Bed.	✗
DDPM [1]	256×256	1k	LSUN Bed.	✗
IF [24]	256×256	1k	LSUN Bed.	✗
SDv2 [2]	768×768	1k	LSUN Bed.	✗
DALL-E2 [3]	1024×1024	0.5k	LSUN Bed.	✗
Midjourney	1024×1024	0.1k	LSUN Bed.	✗
StyleGAN [25]	256×256	0.1k	LSUN Bed.	✗
Dif-SGAN [26]	256×256	0.1k	LSUN Bed.	✗

$$\begin{cases} \chi^n = \chi^{n-1} + \text{Concat}(\psi'_1, \psi'_2, \dots, \psi'_h), \\ \psi^n = \psi^{n-1} + \text{Concat}(\chi'_1, \chi'_2, \dots, \chi'_h). \end{cases} \quad (4)$$

The feature maps are combined through channel concatenation. Ultimately, the adaptive average pooling and a fully-connected layer are utilized for generated-image detection, such as diffusion models and GANs.

E. Loss Function

The proposed DRTNet uses the cross-entropy loss function:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log p(y_i) + (1 - y_i) \cdot \log(1 - p(y_i))], \quad (5)$$

where N denotes the number of samples within a given batch. $y_i \in \{0, 1\}$ represents the true label of each sample. $p(y_i)$ is the probability assigned to the true label y_i of each sample.

IV. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: The real dataset consists of 20,000 uncompressed images sourced from the public ALASKA database, each of resolution of 256×256 , which is widely used in image forensics tasks. The split of ALASKA for training, validation, and testing purposes is 16,000/2000/2000 images, respectively. The generated dataset is DALL-E2 [3], a text-to-image (T2I) database [6]. It encompasses 20,000 images that have been extracted to feature specific spots or tangible objects from the chosen ALASKA images. The split of DALL-E2 for training, validation, and testing purposes is 16,000/2000/2000 images. The training datasets also include the reconstructed images of ALASKA and DALL-E2 using the SDv1 [2] model. The testing dataset is from unseen DiffusionForensics [7] database generated by diffusion models or GANs with two classes: ImageNet [27] and LSUN-Bedroom [28]. LSUN-Bedroom collects bedroom images generated by distinct diffusion models or GANs. Among these, ADM [23], PNDM [17] and DDPM [1] generate images through unconditional diffusion processes. The text-to-image diffusion models include

TABLE II

COMPARISONS OF GENERATED-IMAGE DETECTION ACCURACY (ACC, %), INCLUDING UNSEEN DIFFUSION MODELS AND GANS. THE DALL·E2 (T2I) HAS BEEN TRAINED AND TESTED, WHEREAS IMAGENET AND LSUN BEDROOM ARE UNSEEN, BEING TESTED DIRECTLY WITHOUT PRIOR TRAINING. (THE BEST AND THE SECOND BEST RESULTS ARE BOLD AND UNDERLINED, RESPECTIVELY.)

Source	T2I [6]	ImageNet		LSUN Bedroom									Avg. Acc
Methods	DALL·E2	SDv1	ADM	LDM	PNDM	DDPM	IF	SDv2	DALL·E2	Midjourney	StyleGAN	Dif-SGAN	Acc
F3Net [5]	97.58	85.21	70.36	69.77	52.12	96.68	90.90	85.96	99.90	83.08	64.30	73.11	80.75
DualNet [6]	96.60	98.60	80.51	99.05	49.18	98.89	96.50	98.95	99.30	99.50	97.70	89.26	<u>92.00</u>
DIRE [7]	97.73	88.20	77.25	84.15	58.45	65.97	94.50	94.65	97.30	87.50	57.16	72.80	81.30
AERO. [20]	80.90	98.83	74.68	99.99	60.72	78.70	98.36	89.21	84.64	89.80	73.77	77.46	83.92
ResNet50	79.35	86.36	75.56	84.66	86.46	77.16	84.96	72.56	85.71	86.07	84.21	82.71	82.15
DRTNet	97.12	98.30	81.85	99.50	56.80	99.15	97.90	97.30	99.20	98.50	98.35	98.10	93.51

TABLE III

COMPARISONS OF UNSEEN ROBUSTNESS AGAINST POST-PROCESSING PROCEDURES (ACC, %)

Methods	Chromaticity	Brightness	Contrast	Sharpness	Rotation	Gaussian Blur	JPEG (Q = 95)	Avg. Acc
F3Net [5]	96.13	70.63	70.55	91.47	64.32	62.78	78.87	76.39
DualNet [6]	82.53	77.20	86.28	72.32	68.43	50.00	50.40	69.59
DIRE [7]	80.95	81.05	85.33	77.55	73.80	75.67	82.35	<u>79.53</u>
AEROBLADE [20]	76.96	69.33	70.69	78.54	52.48	79.14	80.67	72.54
ResNet50	76.70	76.75	65.70	77.45	58.67	78.00	79.35	73.23
DRTNet	92.53	85.55	90.95	83.55	80.85	54.87	75.05	80.48

TABLE IV

ABLATION STUDY RESULTS OF DRTNET.

Source	T2I [6]	ImageNet		LSUN Bedroom									Avg. Acc
Methods	DALL·E2	SDv1	ADM	LDM	PNDM	DDPM	IF	SDv2	DALL·E2	Mid.	StyleGAN	Dif-SGAN	Acc
DRTNet	97.12	98.30	81.85	99.50	56.80	99.15	97.90	97.30	99.20	98.50	98.35	98.10	93.51
DRTNet-w/o Rec.	96.90	99.05	80.85	99.25	49.25	99.28	98.20	98.80	99.50	100.00	99.15	85.70	92.16
DRTNet-w/o SRM	96.33	97.85	77.55	99.25	50.05	99.22	95.70	92.25	96.50	96.50	91.85	97.00	90.84
DRTNet-w/o NPR	94.80	94.05	79.30	97.90	47.95	95.18	97.20	92.85	97.90	91.00	95.95	86.95	89.25
DRTNet-w/o DCT	96.30	98.60	80.30	99.15	49.30	98.57	98.00	99.15	98.80	100.00	97.50	96.85	92.71

LDM [2], SDv1 [2], SDv2 [2], IF [24], DALL·E2 [3], Midjourney. The StyleGAN [25] and Diff-StyleGAN [26] generate images through GANs. Tab. I shows the testing datasets of real and generated images used in this work. The “seen” in Tab. I refers to whether the dataset has been pre-trained.

2) *Data Pre-processing and Augmentation*: During training and testing, the input images are resized to a resolution of 256×256 and then center-cropped to 224×224 . To validate the model’s robustness against data post-processing procedures, extensive data augmentations are conducted, including random rotation, Gaussian blur, JPEG compression with 95% quality, chromaticity, brightness, contrast and sharpness adjustments.

3) *Evaluation Metrics*: Following the related works, we adopt detection accuracy (ACC) as the evaluation metric. The threshold for accuracy calculation is set to 0.5.

4) *Baselines*: F3Net (ECCV 2020) [5] utilizes Frequency-aware Image Decomposition (FAD) to learn forgery patterns, and employs Local Frequency Statistics (LFS) to extract high-level semantics. DualNet (APSIPA ASC 2023) [6] contains a residual and a content stream, which employs SRM to obtain texture details and captures forged traces in low frequency. DIRE (ICCV 2023) [7] evaluates the error between a given image and its reconstruction using DDIM. AEROBLADE (CVPR 2024) [20] adopts a training-free strategy for detecting

latent diffusion images using autoencoder reconstruction errors. ResNet50 [19] is a deep residual network with 50 layers.

B. Accuracy of Generated-image Detection

The number of training epochs is 10. We adopt the Adam optimizer with a learning rate of 0.0002. For training, we utilize a NVIDIA A100 GPU. The comparisons of generated-image detection accuracy are shown as Tab. II. AEROBLADE [20] is a training-free method, all tests are unseen. Existing detection methods, such as F3Net [5], DualNet [6], and DIRE [7], have demonstrated near-perfect detection accuracies on seen DALL·E2 (T2I) datasets. However, in real-world scenarios, it is crucial to achieve good accuracies on images generated by unseen models. The performance of these methods on unseen datasets like ImageNet and LSUN Bedroom are not satisfactory. Overall, The proposed DRTNet achieves superior accuracy across most types of generated images, about 93.51%. DRTNet is not only effective in detecting unseen diffusion models, but also generalized across unseen GANs. DualNet is a close second, about 92.00%. F3Net, DIRE, AEROBLADE, and ResNet50 suffer from a decline when detecting unseen generated models, both methods have an average accuracy of approximately 80%-84%. It proves the effectiveness of DRTNet architecture. DRTNet encounters a

limitation in detecting PNDM, potentially due to the diffusion reconstruction model utilizing Stable Diffusion, whereas PNDM is a score-based generative model.

C. Robustness against unseen Post-Processing Procedures

Besides evaluating the performance on unseen generative models, the robustness against unseen perturbations is equally crucial, as images are often affected by diverse degradations in practical scenarios. Therefore, we examine the robustness of baselines and DRTNet in seven degradations, *i.e.*, chromaticity, brightness, contrast and sharpness adjustments, random rotation, Gaussian blur, and JPEG compression (quality = 95). The comparisons of robustness against unseen post-processing procedures are shown as Tab. III. It is worth noting that DualNet [6], AEROBLADE [20], and ResNet50 experience a significant performance drop of approximately 69%-73%, highlighting the lack of robustness in existing detectors. While the proposed DRTNet achieves a satisfactory performance of about 80.48%, demonstrating remarkably robust to unseen post-processing procedures. DIRE [7] also exhibits decent robustness, with an accuracy of approximately 79.53%.

D. Ablation study

To validate the effectiveness of dual-branch inputs and texture feature extraction network, we conduct ablation study by removing the reconstructed images or feature extraction modules. The proposed DRTNet includes the input images $\mathbf{x}$ and reconstructed images $\tilde{\mathbf{x}}$. DRTNet-w/o Rec. only includes the input images $\mathbf{x}_r$ and $\mathbf{x}_g$. DRTNet-w/o SRM only removes the SRM module, similarly DRTNet-w/o NPR and DRTNet-w/o DCT are obtained. The ablation study results is shown as Tab. IV. We find that removing either the reconstructed images $\tilde{\mathbf{x}}$ or any of the feature extraction modules lead to a decrease in detection accuracy.

V. CONCLUSION

In this paper, we propose a novel generated-image detector named DRTNet. The proposed approach leverages the characteristics of the reconstruction error of real and generated images, and a spatial and frequency texture feature extraction. In general, DRTNet consists of a dual-branch inputs including input images and their reconstructed images obtained by Stable Diffusion's VAE to provide comprehensive information. Then the noise patterns, neighboring pixel relationship, and frequency features of the dual-branch inputs are extracted. Subsequently, a dual cross-attention is adopted to fuse the extracted features of the input images and the reconstructed images. Extensive experiments demonstrate that DRTNet is a generalizable and robust generated-image detector, with significant effectiveness in detecting unseen diffusion models and GANs. Notably, the proposed DRTNet shows superior robustness against unseen post-processing procedures.

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